DP Computer Science: A 2.2.1 Network Topologies

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Define and describe different network topologies: **bus, ring, star, tree, mesh, hybrid**
- Analyse the **advantages and disadvantages** of each topology
- Evaluate the **factors influencing network design** (cost, scalability, reliability, etc.)
- Apply knowledge to **recommend suitable topologies** for different settings
- Explain **emerging trends** in network architecture (SDN, cloud computing)

Key Vocabulary

Term	Definition
Network topology	The physical or logical arrangement of devices in a network
Bus topology	All devices connect to a single communication line (cable)
Ring topology	Devices are connected in a closed loop/circle
Star topology	All devices connect to a central hub or switch
Tree topology	A hierarchical structure with a root node and branches
Mesh topology	Every device connects to every other device (fully connected)
Hybrid topology	A combination of two or more topologies
Node	Any device connected to the network (computer, printer, switch, etc.)
Fault tolerance	The ability of a network to continue functioning when parts fail
Scalability	The ability of a network to grow and accommodate more devices

IB ATL Skills

Skill	Description
Thinking	Critical thinking, analysis, problem-solving
Communication	Explaining concepts, presenting information, collaborating in discussions
Research	Gathering information and evaluating sources
Self-Management	Organisation, time management, independent learning

2. Network Topologies Overview

What is a Network Topology?

> **Network topology describes how devices connect in a network—the physical or logical arrangement of nodes and links.**

The choice of topology affects:

- **Cost** (cabling, hardware)
- **Reliability** (fault tolerance)
- **Performance** (speed, collisions)
- **Scalability** (ease of adding devices)
- **Complexity** (installation and maintenance)

Topology Comparison at a Glance

Topology	Diagram	Cost	Reliability	Scalability	Best For
Bus	`─•─•─•─`	Low	Poor	Limited	Small temporary networks

Ring	`•─•`
	`│ │`
	`•─•`
		Medium	Medium	Limited	Small to medium networks

Star	`•─┐`
	`•─┼─•`
	`•─┘`
		Low	Good (hub-dependent)	Good	Homes, small offices

Tree	Hierarchical	Medium	Good	Excellent	Large organisations, campuses
Mesh	Fully connected	High	Excellent	Good	Critical systems, military, internet backbone
Hybrid	Combination	Variable	Variable	Excellent	Large, complex networks

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3. Detailed Topology Descriptions

Bus Topology

Aspect	Description
Structure	All devices connect to a single cable (the "bus")
How it works	Data travels along the cable; all devices see the data but only the intended recipient processes it
Advantages	Simple, inexpensive, easy to set up, requires less cable than other topologies
Disadvantages	Single point of failure (the cable); limited cable length; collisions slow down the network; difficult to troubleshoot
Scalability	Poor—adding devices slows the network
Real-world use	Small temporary networks; legacy networks; simple setups

Diagram:

```text
    ┌────────┐  ┌────────┐  ┌────────┐
    │ Device │  │ Device │  │ Device │
    └───┬────┘  └───┬────┘  └───┬────┘
```

```
    │        │        │
    └────────┼────────┘
             │
        ┌────┴────┐
        │  Cable  │ (The "Bus")
        └─────────┘
```

Analogy: A bus topology is like a **shared hallway**—everyone shares the same space, and messages are passed along until they reach the right person.

Ring Topology

Aspect	Description
Structure	Devices are connected in a closed loop/circle
How it works	Data travels in one direction (or both) around the ring; each device passes data to the next
Advantages	Ordered transmission (no collisions); predictable data flow; simple cabling
Disadvantages	One failed device can break the whole network; difficult to troubleshoot; adding devices requires network downtime
Scalability	Limited—large rings can become slow
Real-world use	Small to medium networks; legacy networks (Token Ring); SONET/SDH networks

Diagram:

```text
       ┌────────┐
       │ Device │
       └────┬───┘
            │
   ┌────────┼────────┐
   │        │        │
   │   ┌────┴───┐    │
   │   │ Device │    │
   │   └────┬───┘    │
   │        │        │
   │   ┌────┴───┐    │
   │   │ Device │    │
   │   └────┬───┘    │
   └────────┼────────┘
            │
       ┌────┴───┐
       │ Device │
       └────────┘
```

Analogy: A ring topology is like a **round-robin meeting**—each person gets a turn to speak, and messages are passed around the circle.

Star Topology

Aspect	Description
Structure	All devices connect to a central hub or switch
How it works	Data travels from device → central hub → destination device
Advantages	Easy to install and manage; fault-tolerant (one cable failure doesn't affect others); easy to add devices; good performance (no collisions with switches)
Disadvantages	Central hub is a single point of failure; requires more cabling than bus
Scalability	Good—add devices by connecting to the hub
Real-world use	Home networks, small offices, school computer labs

Diagram:

```text
                ┌────────┐
                │ Device │
                └───┬────┘
                    │
                ┌───┴────┐
                │ Device │
                └───┬────┘
                    │
    ┌────────┐  ┌───┴────┐  ┌────────┐
    │ Device ├──┤  HUB/  ├──┤ Device │
    └────────┘  │ SWITCH │  └────────┘
                └───┬────┘
                    │
                ┌───┴────┐
                │ Device │
                └────────┘
```

Analogy: A star topology is like a **school office**—all messages go through the office, which distributes them to the correct person.

Tree Topology

Aspect	Description
Structure	A hierarchical arrangement with a root node and branches (combines star and bus)
How it works	Data travels from leaf nodes up through the hierarchy to the root, then down to other branches
Advantages	Scalable; hierarchical management; good performance; fault-tolerant if designed with redundancy
Disadvantages	Root node is critical; more complex to design; more cabling than simple topologies
Scalability	Excellent—add branches as needed
Real-world use	Large organisations, university campuses, corporate networks

Diagram:

```text
           ┌──────────┐
           │  ROOT    │
           │  NODE    │
           └────┬─────┘
                │
     ┌──────────┼──────────┐
     │          │          │
┌────┴─────┐ ┌──┴──────┐ ┌─┴───────┐
│ Branch   │ │Branch   │ │Branch   │
└────┬─────┘ └──┬──────┘ └────┬────┘
     │          │             │
┌────┴─────┐    │        ┌────┴─────┐
│ Device   │    │        │ Device   │
└──────────┘    │        └──────────┘
                │
           ┌────┴─────┐
           │ Device   │
           └──────────┘
```

Analogy: A tree topology is like a **company org chart**—information flows from the CEO down through departments to individual employees.

Mesh Topology

Aspect	Description
Structure	Every device connects to every other device (fully connected)
How it works	Multiple paths between any two devices; data can choose alternative routes
Advantages	Extremely reliable; fault-tolerant (multiple paths); redundant; no single point of failure
Disadvantages	Expensive; complex to install; requires significant cabling; difficult to manage
Scalability	Limited—each new device requires connections to all others (n(n-1)/2 connections)
Real-world use	Military communication systems; critical infrastructure; internet backbone

Diagram (Full Mesh with 5 devices):

```text
┌───── Device A ─────┐
│    │     │     │    │
│    │     │     │    │
│  ┌─┴─────┼─────┴┐  │
│  │       │      │ │
│  │  Device B ── Device C
│  │       │      │ │
│  └─┬─────┼──────┘ │
│    │     │     │    │
└────┴── Device D ───┘
           │     │
     ┌─────┴─────┘
     │
Device E
```

Analogy: A mesh topology is like a **city road network**—there are multiple routes to get from one place to another, so if one road is blocked, you can take another.

Hybrid Topology

Aspect	Description
Structure	A combination of two or more topologies
How it works	Different parts of the network use different topologies based on their needs
Advantages	Flexible; customisable; can meet diverse requirements; combines strengths of multiple topologies
Disadvantages	Complex to design; expensive; difficult to manage
Scalability	Excellent—different areas can be scaled independently
Real-world use	Large enterprise networks; university campuses; ISP networks

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4. Factors Influencing Topology Choice

Factor	Description	How It Affects Choice
Cost	Budget for hardware, cabling, installation	Lower cost → bus or star; higher cost → mesh or hybrid
Reliability	Importance of network uptime	High reliability → mesh or star (with redundant hub)
Scalability	Need to add more devices	High scalability → tree, star, or hybrid
Speed/Performance	Data transfer requirements	Star or mesh (switches reduce collisions)
Complexity	Ease of installation and management	Low complexity → star or bus; high complexity → mesh or hybrid
Physical Layout	Building size, geography	Large area → tree or hybrid; small area → star
Fault Tolerance	Ability to survive failures	High fault tolerance → mesh or ring (with dual-ring)

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5. Topologies in Different Settings

Setting	Recommended Topology	Why?
Home Network	**Star**	Easy to set up, affordable, suitable for small numbers of devices

Setting	Recommended Topology	Why?
Small Office	Star or Hybrid	Easy to manage; can add devices easily; can add backup hub for redundancy
University Campus	Hybrid (Star + Tree)	Connects multiple buildings and departments; scalable; hierarchical management
Military Communication	Mesh	Highest reliability and fault tolerance; survives damage
Airplane Internal Network	Hybrid	Critical systems need redundancy (mesh); entertainment may use star/bus
Internet Backbone	Mesh	High redundancy; handles massive traffic; multiple paths
Smart Home	Mesh (wireless)	Reliable coverage throughout the house; self-healing
Rural Community Wi-Fi	Mesh (wireless)	Extends network range; shares internet connection across households

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6. Emerging Trends in Network Architecture

Trend	Description
SDN (Software-Defined Networking)	Separates network control from data forwarding; makes networks more flexible and programmable
Cloud Computing	Networks increasingly rely on cloud services; topology must support cloud connectivity
Wireless Mesh Networks	Self-configuring, self-healing wireless networks; growing in popularity for smart homes and communities
IoT (Internet of Things)	Massive numbers of devices connected; requires scalable, efficient topologies
5G Networks	High-speed, low-latency mobile networks; uses complex hybrid topologies

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7. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Engage	10 min	"How does your phone connect to the internet?"; define network topology

Phase	Time	Activity
Interactive Activity	15 min	Think-Pair-Share: Match scenarios to topologies; justify choices
Explain	15 min	Core topologies (bus, star, mesh); factors influencing design
Elaborate	15 min	Additional topologies (ring, tree, hybrid); emerging trends
Wrap-up	5 min	Key takeaways; exit ticket

Activity: Network Topology Match-Up

Instructions: Students match each scenario to the most suitable topology and justify their choice.

Scenario	Best Topology	Justification
Home Network	Star	Easy to set up, affordable, suitable for small networks
Small Office	Star or Hybrid	Reliable; easy to manage; can add devices as needed
University Campus	Hybrid (Star + Tree)	Connects multiple buildings; scalable; hierarchical management
Military Communication	Mesh	Highest reliability and fault tolerance
Airplane Internal Network	Hybrid	Critical systems need redundancy; entertainment less critical
Internet Backbone	Mesh	Extremely robust; handles massive traffic
Smart Home	Mesh	Reliable coverage; self-healing
Rural Community Wi-Fi	Mesh	Extends range; shares internet connection

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8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Network Topology Match-Up	Assess ability to match scenarios to topologies and justify choices
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is a network topology?
2. What is a disadvantage of a bus topology?
3. Why is a mesh topology used for critical systems?
4. Which topology is most common in home networks and why?

9. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and real-world examples
Challenge	Encourage advanced students to research SDN, 5G, or IoT network architectures

10. Resources

- **Presentation** (Slide Deck)
- **eBook** (A2.2.1 with quiz)
- **Quizlet** (A2.2.1 vocabulary flashcards)
- **Activity:** Network Topology Match-Up handout

11. Summary: Key Takeaways

Topology	Key Feature	Best For
Bus	Simple, cheap, single cable	Small temporary networks
Ring	Closed loop, ordered data flow	Small to medium networks
Star	Central hub, easy to manage	Homes, small offices
Tree	Hierarchical, scalable	Large organisations, campuses
Mesh	Fully connected, fault-tolerant	Critical systems, internet backbone
Hybrid	Combination, flexible	Large complex networks

> "*The best topology depends on the specific needs—cost, reliability, scalability, and fault tolerance all play a role in the decision.*"